The Geography of R&D and the Onshoring of Innovation: Evidence from TCJA Section 174

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Section 174 and the Tax Treatment of R&D

- TCJA (2017) was primarily deficit financed
- A pay-for: §174 in TCJA replaces expensing of R&D with amortization
- §174 set to go into effect for tax years beginning after **Dec. 31, 2021**
  - Business community expected repeal during each Congress 2018–2021 (Cowx et al. 2025)
- Extra §174 goal: onshoring innovative activity by making domestic R&D cheaper than foreign R&D
  - **Domestic R&D:** 5-year amortization $\implies$ after-tax cost of R&D $\uparrow\approx$ 5.4%
  - **Foreign R&D:** 15-year amortization $\implies$ after-tax cost of R&D $\uparrow\approx$ 12.4%
  - (21% tax rate, 10% discount rate)

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This paper

Exploiting cross-firm heterogeneity in the *magnitude* of the tax shock, driven by pre-TCJA domestic vs. foreign R&D mix

Motivation and Research Question

Three big-picture questions:

1. **Policy impact:** Did the 5 vs. 15-year wedge achieve its onshoring goal?
2. **Elasticity:** How elastic are innovation outputs to the user cost of R&D?
3. **Geographic substitutability:** Can domestic and foreign R&D inputs substitute for one another?

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Research Question

How did the **quantity, location, and characteristics** of innovation outputs at U.S. firms respond to an exogenous, firm-specific shock to the **user cost of R&D**?

Empirical Design: Three Measurement Challenges

$$Y_{igt} = \alpha_i + \gamma_{jt} + \beta \left(T_i \times Post_t \right) + \theta' W_{i,t-1} + \varepsilon_{igt}$$

Problem 1: Unobserved internal R&D allocation (domestic vs. foreign)

- Necessary to define treatment status

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Problem 1: Unobserved internal R&D allocation (domestic vs. foreign)

- Necessary to define treatment status
- *Solution:* Pre-TCJA inventor location data (PatentsView, 2010–2016)

$$\text{share}_i = \frac{\#\text{ international inventors}_{i,\,2010\text{–}2016}}{\#\text{ total inventors}_{i,\,2010\text{–}2016}}, \qquad T_i = 1[\text{share}_i > \text{median}]$$

- *Validation:* mean US-inventor share is 77%, close to the 81% domestic R&D share in confidential Census data (Glaeser et al. 2025)

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Problem 2: R&D allocation also unobservable as an outcome

- To measure impact of shock on innovative activity in places $\tilde{g} \in \{US, \text{ Foreign}\}$

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Problem 2: R&D allocation also unobservable as an outcome

- To measure impact of shock on innovative activity in places $\tilde{g} \in \{US,$ Foreign$\}$
- *Solution:* Measure allocation of invention outputs (patents) by inventor location
  - $Y_{i,g,t}$ = outcome Y at firm i, for inventions of type g, in year t
  - $g \in$ {Foreign-only, US-only, Collaborative}, where "Collaborative" means at least one domestic and one foreign inventor on the same patent
  - Y: count of distinct inventors, patent applications, and technology classes (CPC subclasses, "breadth")

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Problem 3: Need output data through 2024

- Grants have multi-year lag; ≈50% of 2024 innovations (applications) are not granted yet

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Problem 3: Need output data through 2024

- Grants have multi-year lag; ≈50% of 2024 innovations (applications) are not granted yet
- *Solution:* Use data on patent *applications*
  - ⇒ All 2024 applications were published by July 2025
  - Hurdle: Some data on applications requires scraping Google Patents
- Restriction: Hard to assess changes in "patent quality"
  - Forward citations and patent-value measures are defined on granted patents, so not available for applications

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Bonus Problem 4: Y has zeros

- $Y_{i,g,t}$: count of distinct inventors, patent applications, technology classes
- These will contain zeros, especially for the foreign and collaborative categories
- Following Cohn, Liu & Wardlaw (2022): Poisson PML for count outcomes
- Will also show log1p and IHS specifications as checks, for ease of interpretation; their estimates depend on the units of Y (Chen & Roth 2024)

Sample Construction and Summary Statistics

Sample construction

- Compustat (CCM) fiscal years 2019–2024, balanced sample (6 years each)
- Exclude utilities and financials (SIC 4900–4999, 6000–6999)
- Link PatentsView patent applications to PERMNO/GVKEY via KPSS crosswalk
- Formation period (2010–2016, granted patents): 5+ inventor credits
- Active innovators: 1+ granted patent filed in 2019–2021
- $T_i = 1$ if formation period foreign-inventor share is above the median
- Final sample: 801 firms (496 control, 305 treated); 4,680 firm-years

Sample Construction and Summary Statistics

Panel A: Summary statistics

	N	Mean	SD	P10	P50	P90
T_i	4,680	0.381	0.486	0.000	0.000	1.000
Firm-level Outcomes:						
R&D	4,680	0.098	0.150	0.000	0.046	0.255
CAPX	4,680	0.030	0.030	0.003	0.021	0.066
Inventors	4,680	81.494	343.649	0.000	4.000	158.500
Financials:						
Log(Assets)	4,676	7.866	2.263	4.839	7.809	10.926
MB	4,675	2.888	2.483	1.057	2.069	5.695
Cash/Assets	4,676	0.263	0.245	0.032	0.169	0.665

Sample Construction and Summary Statistics

Panel B: By treatment group

	Control (T=0)	Treat (T=1)	Diff	SE
Firm-level Outcomes:				
R&D	0.102	0.090	0.012***	(0.005)
CAPX	0.029	0.032	-0.003***	(0.001)
Inventors	46.672	138.175	-91.502***	(10.260)
Financials:				
Log(Assets)	7.524	8.423	-0.899***	(0.067)
MB	2.989	2.724	0.265***	(0.075)
Cash/Assets	0.279	0.237	0.042***	(0.007)

Treated firms are larger and more inventor-intensive $\Rightarrow$ firm FE and industry$\times$year FE

First Stage: R&D Falls Where the Foreign Penalty Bites

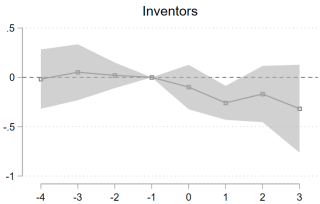
	R&D		CAPX		Inventors	
	(1)	(2)	(3)	(4)	(5)	(6)
$T_i \times Post_t$	-0.013**	-0.013**	0.001	0.001	-0.169**	-0.139**
	(0.006)	(0.006)	(0.001)	(0.001)	(0.072)	(0.071)
Log(Assets)		-0.006		0.001		0.250**
		(0.008)		(0.001)		(0.122)
MB		-0.003		0.001*		-0.034*
		(0.002)		(0.000)		(0.018)
Cash/Assets		0.009		0.011**		-0.714**
		(0.040)		(0.005)		(0.317)
Fixed effects:						
Firm	Yes	Yes	Yes	Yes	Yes	Yes
Ind x Year	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	4,623	4,618	4,623	4,618	3,538	3,534

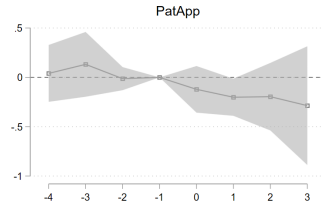
- R&D/assets falls by ≈1.3pp at treated firms (≈13% of the sample mean of 9.8%)
- Placebo: no differential change in CAPX ⇒ not a general investment contraction
- Total inventor headcount down ≈13–16% (Poisson)
- → the shock reaches real inventive activity; decompose by inventor geography next

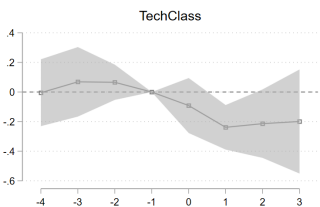
The Innovative Activities of Foreign Inventors Fell

	Inventors		PatApp		TechClasses	
	(1)	(2)	(3)	(4)	(5)	(6)
$T_i \times Post_t$	-0.245***	-0.202**	-0.187*	-0.103	-0.194**	-0.153*
	(0.094)	(0.098)	(0.111)	(0.120)	(0.077)	(0.078)
Log(Assets)		0.364**		0.809***		0.417**
		(0.174)		(0.251)		(0.172)
MB		-0.081***		-0.092**		-0.070**
		(0.026)		(0.037)		(0.029)
Cash/Assets		-0.697		-1.233		-0.540*
		(0.488)		(1.110)		(0.282)
Fixed effects:						
Firm	Yes	Yes	Yes	Yes	Yes	Yes
Ind x Year	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	2,740	2,739	2,750	2,749	2,252	2,252

The Innovative Activities of Foreign Inventors Fell







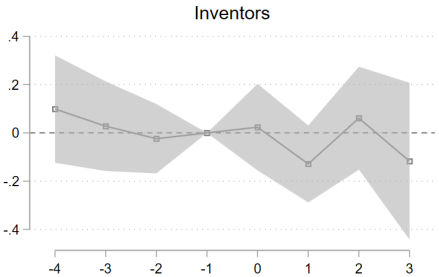
- 2017–2018 added to figures to show pre-trends
- Event time 1 = FY2022; −1 (FY2020) omitted
- Pre-trends flat; sharp break after 2022

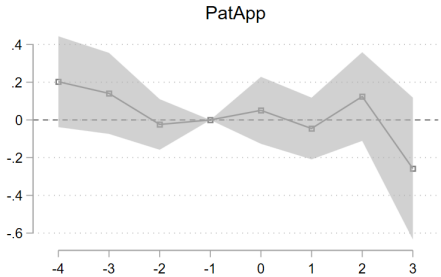
Domestic Innovative Activity Did Not Offset the Decline

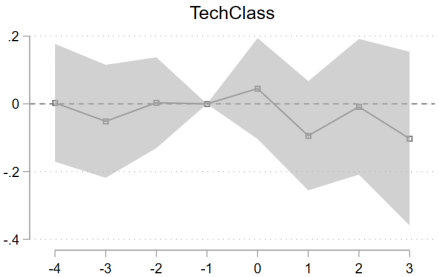
	Inventors		PatApp		TechClasses	
	(1)	(2)	(3)	(4)	(5)	(6)
$T_i \times Post_t$	-0.073	-0.052	-0.055	-0.029	-0.066	-0.050
	(0.073)	(0.070)	(0.082)	(0.080)	(0.064)	(0.065)
Log(Assets)		0.143		0.172*		0.119
		(0.101)		(0.100)		(0.081)
MB		0.000		0.004		0.014
		(0.014)		(0.014)		(0.014)
Cash/Assets		-0.665**		-0.718**		-0.321
		(0.300)		(0.301)		(0.239)
Fixed effects:						
Firm	Yes	Yes	Yes	Yes	Yes	Yes
Ind x Year	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	3,277	3,273	3,277	3,273	3,143	3,139

Upper 95% CI: rules out a domestic rise of more than $\sim$7–9% in inventors (vs. -18 to -22% for international)

Domestic Innovative Activity Did Not Offset the Decline







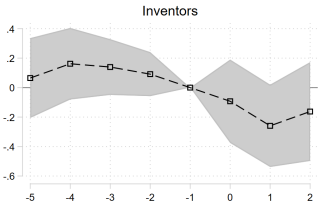
Within-Firm Triple Difference: International vs. Domestic

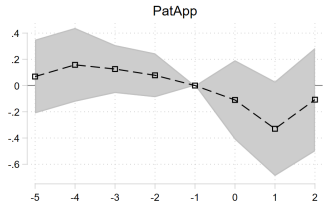
	Inventors		PatApp		TechClasses	
	(1)	(2)	(3)	(4)	(5)	(6)
$T_i \times Post_t \times Intl_g$	-0.202**	-0.239**	-0.194*	-0.254*	-0.144	-0.182**
	(0.101)	(0.113)	(0.117)	(0.138)	(0.088)	(0.092)
$T_i \times Post_t$	-0.045		-0.020		-0.059	
	(0.072)		(0.083)		(0.060)	
Fixed effects:						
Firm x Group	Yes	Yes	Yes	Yes	Yes	Yes
Ind x Year x Group	Yes	Yes	Yes	Yes	Yes	Yes
Firm x Year	No	Yes	No	Yes	No	Yes
Obs.	6,055	4,264	6,065	4,282	5,421	3,364

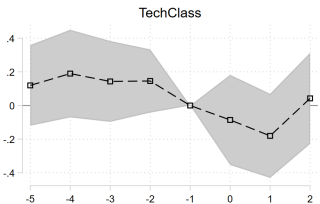
Firm×year FE absorb all firm-level shocks;
Stacked international and domestic outcomes;

$T_i \times Post_t \times Intl_g$ = extra effect on international activity.

Within-Firm Triple Difference: International vs. Domestic





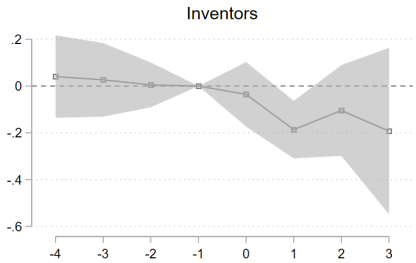


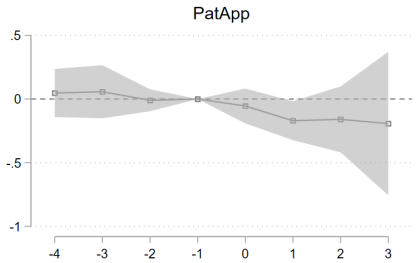
- Coefficients on $T_i \times Intl_g \times$ event year
- Year-1 dip on all three outcomes, partly reverting in year 2
- Pre-period points are positive but jointly insignificant ($p = 0.57$ inventors, 0.69 applications, 0.53 technology classes)

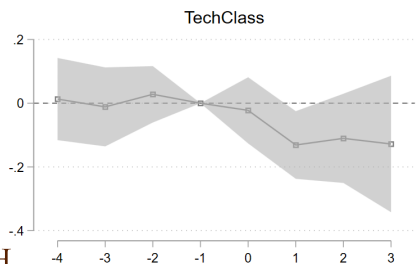
Overall Innovation Outputs Decline

	Inventors		PatApp		TechClasses	
	(1)	(2)	(3)	(4)	(5)	(6)
$T_i \times Post_t$	-0.169**	-0.139**	-0.164	-0.093	-0.108**	-0.084*
	(0.072)	(0.071)	(0.107)	(0.112)	(0.047)	(0.048)
Log(Assets)		0.250**		0.621***		0.282**
		(0.122)		(0.201)		(0.124)
MB		-0.034*		-0.064***		-0.027
		(0.018)		(0.024)		(0.023)
Cash/Assets		-0.714**		-0.994		-0.317*
		(0.317)		(0.633)		(0.192)
Fixed effects:						
Firm	Yes	Yes	Yes	Yes	Yes	Yes
Ind x Year	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	3,538	3,534	3,538	3,534	3,538	3,534

Overall Innovation Outputs Decline







Robustness: Estimator and Functional Form

	Inventors		PatApp		TechClasses	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: International-only teams						
Quantile	-0.259**	-0.244**	-0.234	-0.208	-0.261***	-0.248***
$f(y) = y/(\bar{y}_1^{pre} - \bar{y}_0^{pre})$	(0.117)	(0.113)	(0.176)	(0.187)	(0.061)	(0.066)
OLS	-0.178***	-0.168***	-0.146***	-0.134***	-0.195***	-0.186***
$f(y) = log(y+1)$	(0.053)	(0.053)	(0.046)	(0.046)	(0.041)	(0.041)
OLS	-0.208***	-0.197***	-0.170***	-0.156***	-0.234***	-0.224***
$f(y) = asinh(y)$	(0.063)	(0.063)	(0.054)	(0.054)	(0.050)	(0.050)
Panel B: Domestic-only teams						
Quantile	-0.166	-0.146	-0.152	-0.139	-0.293	-0.267
$f(y) = y/(\bar{y}_1^{pre} - \bar{y}_0^{pre})$	(0.247)	(0.255)	(0.211)	(0.218)	(0.193)	(0.192)
OLS	-0.044	-0.031	-0.059	-0.050	-0.036	-0.028
$f(y) = log(y+1)$	(0.062)	(0.062)	(0.051)	(0.051)	(0.046)	(0.046)
OLS	-0.042	-0.026	-0.064	-0.054	-0.034	-0.024
$f(y) = asinh(y)$	(0.073)	(0.074)	(0.061)	(0.061)	(0.057)	(0.057)

Which margin(s) matter: Productivity, Relocation, Attrition?

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- Consistent with endogenous culling of lower-output foreign projects at the extensive margin
- Asymmetry (foreign quantity ↓, productivity ↑; domestic flat) is hard to square with a general financial shock

	PatApp per Intl Inventor		PatApp per US Inventor	
	(1)	(2)	(3)	(4)
$T_i \times Post_t$	0.308**	0.319**	0.006	0.020
	(0.137)	(0.136)	(0.131)	(0.127)
Log(Assets)		0.107		0.100
		(0.141)		(0.136)
MB		0.004		0.028
		(0.024)		(0.022)
Cash/Assets		-0.481		-0.291
		(0.370)		(0.418)
Fixed effects:				
Firm	Yes	Yes	Yes	Yes
Ind x Year	Yes	Yes	Yes	Yes
Obs.	2,447	2,447	3,100	3,096

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- No significant evidence of:

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- Asymmetry (foreign quantity ↓, productivity ↑; domestic flat) is hard to square with a general financial shock
- No significant evidence of:
  - Relocation: 0.5% of foreign-located inventors later appear in the US
  - US inventor exit or entrance changing across treatment/control

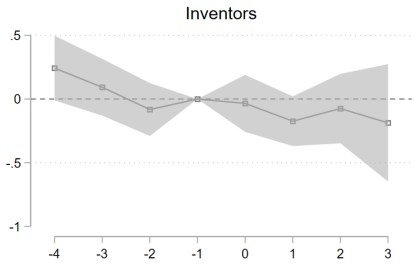
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Obs.	2,447	2,447	3,100	3,096

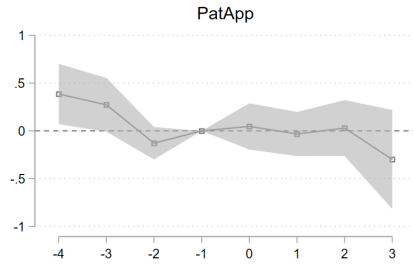
Early Evidence on Spillovers: Collaborative Projects Declined

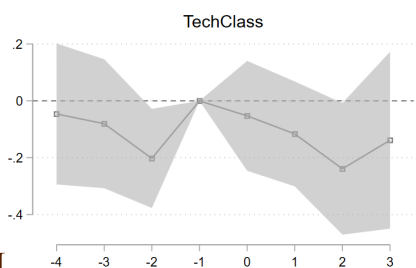
	Inventors		PatApp		TechClasses	
	(1)	(2)	(3)	(4)	(5)	(6)
$T_i \times Post_t$	-0.151*	-0.134*	-0.089	-0.065	-0.111*	-0.091
	(0.081)	(0.078)	(0.091)	(0.089)	(0.065)	(0.065)
Log(Assets)		0.173		0.273**		0.286***
		(0.127)		(0.113)		(0.085)
MB		-0.022		-0.005		-0.008
		(0.022)		(0.018)		(0.022)
Cash/Assets		-0.928**		-0.780*		-0.272
		(0.404)		(0.433)		(0.290)
Fixed effects:						
Firm	Yes	Yes	Yes	Yes	Yes	Yes
Ind x Year	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	2,242	2,241	2,242	2,241	2,242	2,241

Inventors -14% / -12%: roughly half the international-only effect; marginal significance ($p < 0.10$)

Early Evidence on Spillovers: Collaborative Projects Declined







Alternative Explanations: China Decoupling and CHIPS

	Inventors (ex. China)		PatApps (ex. China)		TechClasses (ex. China)	
	(1)	(2)	(3)	(4)	(5)	(6)
$T_i \times Post_t$	-0.251***	-0.206**	-0.212*	-0.116	-0.177**	-0.131
	(0.096)	(0.100)	(0.114)	(0.126)	(0.080)	(0.081)
Log(Assets)		0.361**		0.837***		0.422**
		(0.174)		(0.252)		(0.167)
MB		-0.083***		-0.101***		-0.077***
		(0.026)		(0.038)		(0.028)
Cash/Assets		-0.679		-1.215		-0.562*
		(0.511)		(1.199)		(0.287)
Fixed effects:						
Firm	Yes	Yes	Yes	Yes	Yes	Yes
Ind x Year	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	2,691	2,690	2,691	2,690	2,240	2,240

- **China:** drop any patent with a China-located inventor from the treatment measure and the outcomes ⇒ international effects essentially unchanged (inventors -0.251 / -0.206 vs. -0.245 / -0.202)
- **CHIPS Act:** subsidizes domestic inventors at treated firms ⇒ would work against our domestic null (null is conservative); CAPX placebo is flat

From Reduced Form to a Parameter: Why a Model?

What we found (Poisson DiD on inventors, no controls):

- International-only: $\approx -22\%$ Collaborative: $\approx -14\%$ Domestic-only: $\approx -7\%$ (n.s.)
- Pattern is consistent with domestic–foreign complementarity

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Scope

A stylized patent-production model used as a *disciplining device*: a calibrated map from reduced-form moments to σ, not a full structural estimation

Model: Patent Production with Domestic and Foreign Inventors

Firm $i \in \{T, C\}$ with pre-shock effort $(\bar{e}_d, \bar{e}_f)$ chooses (e_d, e_f):

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$$\max_{e_d, e_f} A_i \left[\alpha_i (e_d - c_d)^\rho + (1 - \alpha_i)(e_f - c_f)^\rho\right]^{\gamma/\rho} - (1 + \tau_d) e_d - (1 + \tau_f) e_f - \frac{\lambda}{2\bar{e}_d}(e_d - \bar{e}_d)^2 - \frac{\lambda}{2\bar{e}_f}(e_f - \bar{e}_f)^2$$

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Production

CES in *discretionary* effort $e_j - c_j$ (Stone–Geary minimum thresholds)

Without c_j, a homothetic CES gives *zero* DiD by construction - both types of inventors respond proportionally

With c_j, treated firms (large foreign effort) have more discretionary margin to re-optimize than control firms

$\sigma = 1/(1-\rho)$: substitution between domestic and foreign effort

γ: returns to scale

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Tax wedge

Effort proxied by R&D spend; §174 amortization raises per-unit cost

$\tau_d = 0.055$, $\tau_f = 0.11$ (exact: 5.4%, 12.4%)

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Adjustment costs

Proportional quadratic (Hayashi-style), scaled by $1/\bar{e}_j$

λ: curvature

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Mapping the model to the DiD estimates

Patents $\propto$ effort $\Rightarrow$ the model's change in $\log e_j$ is its predicted change in j-type patents (log points $\times 100$):

$$\beta_j = \left(\log e_j^T - \log \bar{e}_j^T\right) - \left(\log e_j^C - \log \bar{e}_j^C\right), \quad j \in \{d, f\}$$

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Model (depends on σ)

$$\beta_d(\sigma), \quad \beta_f(\sigma)$$

$\Longleftrightarrow$

Data (patent-application Poisson DiD)

$$\hat{\beta}_d = -5.5, \quad \hat{\beta}_f = -18.7$$

Calibration: Matching the Domestic and Foreign Moments

Pre-shock effort (R&D/assets $\times$ pre-TCJA inventor share)

	Domestic $\bar{e}_d$	Foreign $\bar{e}_f$
Treated	4.0	5.0
Control	9.0	0.4

- Thresholds $c_d = c_f = 0.3$ (10th pctile of pre-shock effort)
- Treatment and control start with different allocations; observed mix treated as an interior optimum

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Estimation

- Target: $\hat{\beta}_f = -18.7$, $\hat{\beta}_d = -5.5$ (patent-application Poisson DiD, no controls)
- Grid over (γ, λ); for each, choose $\sigma \in (0, 7]$ minimizing distance to the empirical pair
- Best fit: $\gamma = 0.76$, $\lambda = 0.16$
- Predicted: $\beta_d = -5.49$, $\beta_f = -18.7$

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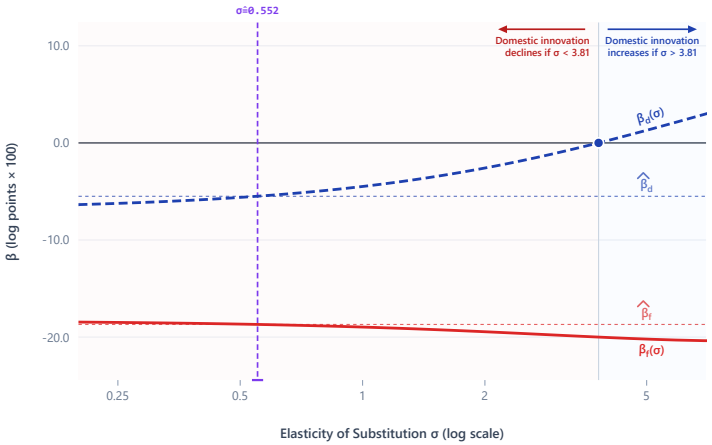
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Result

$\hat{\sigma} = 0.552$: domestic and foreign inventive effort are **complements**

Onshoring Requires $\sigma > 3.8$; the Data Say $\hat{\sigma} \approx 0.55$



- Simulated DiD vs. σ, other parameters fixed
- Foreign effect ≈ -19 to -20 log points for any σ
- Domestic effect turns positive only above $\sigma = 3.81$
- $\hat{\sigma} = 0.55 \ll 3.8$: policy design could not onshore
- Every (γ, λ) with fit error < 0.2 gives $\sigma < 0.85$

Heterogeneity: Domestic Effect by Industry Complementarity

- Model prediction: domestic effect more negative where σ is low (complements)

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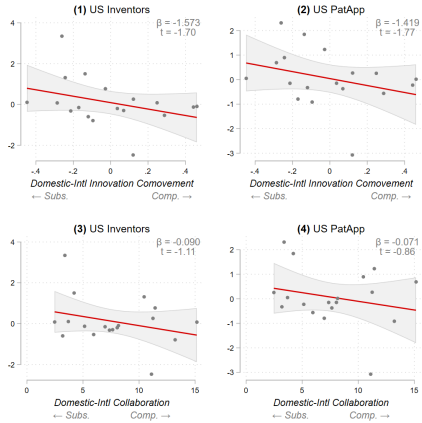
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- Industry-level complementarity proxies, from the pre-period:
  - Co-movement of domestic and international patenting
  - Share of collaborative patents
- Estimate industry-specific DiD
- All four slopes negative, as predicted; top row $|t| \approx 1.7$–1.8, bottom row insignificant
- **Suggestive**, low power



Model Implications and Caveats

OBBBA (2025)

- Restored immediate expensing for domestic R&D (tax years after Dec. 31, 2024)
- Foreign R&D: still 15-year amortization
- Wedge between domestic and foreign R&D *widens*

Caveats

- Calibrated, not estimated: a single σ from pooled DiD moments
- Effort proxied by R&D spend; patents $\propto$ effort
- No adjustment-cost complementarities across locations
  - If disrupting foreign teams raises domestic costs, true σ may be even lower

Contributions

- **1/3: Location of Inventive Activity**
  How the *geography* of innovation (not just its level) responds to a tax-induced user-cost wedge, using inventor-level patent data separated into domestic-only, collaborative, and international teams.

  Location of R&D: Wilson (2009); Moretti & Wilson (2017); Akcigit, Baslandze & Stantcheva (2016); Knoll et al. (2021); Huang, Krull & Ziedonis (2020) · §174 levels: Cowx et al. (2025); Du & Li (2025)

Contributions

- **2/3: Domestic vs. Foreign R&D Substitutability**
  The domestic null disciplines the elasticity of substitution: $\hat{\sigma} \approx 0.55$, far below the $\sigma > 3.8$ onshoring requires. Mirror image of Glennon (2024), where raising the cost of the domestic input shifted work abroad.

  Capital and labor: Desai, Foley & Hines (2009); Kovak, Oldenski & Sly (2021); Chodorow-Reich et al. (2024); Harrison & McMillan (2011) · Inventive inputs: Bilir & Morales (2020); Abramovsky, Griffith & Miller (2017)

Contributions

- **3/3: Unintended Consequences for Cross-Border Teams**
  When innovation is organized through cross-border teams, taxing the foreign input contracts joint knowledge production without a domestic offset.

  Kerr & Kerr (2018); Jaravel, Petkova & Bell (2018); Jones (2009); Foley & Kerr (2013); Huang, Krull & Ziedonis (2020)

Conclusions: The Cost of Onshoring R&D

The Policy Goal: TCJA §174 aimed to onshore R&D by creating a user-cost wedge between domestic (5-year) and foreign (15-year) amortization.

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Thank you!
Questions?

`https://bowen.finance`